Types in Scale: one map

Scale uses several type-like notions. They are not interchangeable and they do not share a representation. This page is the cross-reference; the detailed contract for each remains in its own document.

Notion	Representation	Authority	Consumers	Boundary
Expression inference (P4)	`TypeInference.Type`: variables, primitives, sorts, functions with latent effects; schemes and constraints	`components/typing/type-inference.fs`	Expression elaborator, `--prose-typecheck`, the lowering handoff	Host-side and compile-time. It consumes `ExpressionAst`, not the image or runtime terms. Exact effect sets; no rows, subtyping, or higher-order type unification.
Verification lattice (D25)	`TypeLattice.Lattice`: named fields, field types, finite coordinates or symbolic domains, narrowing guards	`components/typing/type-lattice.fs`	Z3 encoding, Hedgehog generators, the refiner	A projection of a `.scale` domain, not a second source of truth. It is not the runtime set-theoretic sort record; bridging them is a separate decision.
Runtime sorts and value sets	Image-resident `ValueSet`: an optional name plus an extension; declared domains live in `DeclarationsRoot`	`components/prelude/evaluator/value-sets.fs` and the evaluator's set words	Evaluator, set operations, dispatch, sort constructors	Runtime data. The image is the authority; lookup is latest-write-wins and a declared domain only grows. It is not a static type.
Document field types	`ScaleFile.FieldType` and `DomainExpression.FieldType`: boolean, integer, natural, real	`components/core/scale-file.fs` and `components/typing/domain-expression.fs`	The `.scale` parser, the verifier, the F# emitter, the lattice	The editable specification (D7/D9). It is the source the lattice projects; it is not a runtime sort.
Term types	`TermIr.Term` and the portable term document	`components/kernel/term-ir.fs` and `components/kernel/term-portable.fs`	The kernel bridge, the term unifier, the structural verifier	A representation of code as data, not a static type of a value. The portable document is versioned separately.

Rule of thumb

1. A fact about a word's call shape at elaboration time is **expression inference**.
2. A fact about the values a domain field may take is the **verification lattice**.
3. A fact about the declared sort of a runtime value is the **image-resident sort record**.
4. A fact declared in a `.scale` document is a **document field type**.

5. A fact about code as data is a **term**.

None of these converts into another implicitly. A feature that needs a bridge states it in a decision record and names the direction of the conversion.

See also

- `docs/type-inference.typ` — the P4 contract and the single word index.
- `docs/type-lattice.typ` — the verification coordinate system.
- `docs/set-theoretic-typing.typ` — the structural/nominal dial and the runtime sort record.
- `docs/cell-system.typ` — slots, classes, and instances.
- `docs/term-ir.typ` — code as data and the portable term document.